20A 1 ORGANIC ANALYSIS

SPECIFICATION REFERENCE 20.1

LEARNING OBJECTIVES

- Be able to deduce the empirical formulae, molecular formulae and structural formulae from data drawn from combustion analysis, element percentage composition, characteristic reactions of functional groups, infrared spectra, mass spectra and NMR spectra (both ^{13}C and proton).

TRADITIONAL METHODS

The traditional method of analysis is to find the empirical formula, followed by the molecular formula and then the functional groups present.

FINDING THE EMPIRICAL AND MOLECULAR FORMULAE

You learned the method for finding the empirical formula of a compound in **Topic 1 (Book 1: IAS)**.

To find the empirical formula from the percentage composition by mass of the elements you must first divide the percentage composition of each element by its relative atomic mass. You then divide the values obtained by the smallest value. The numbers you obtain should give you an obvious whole number ratio.

This worked example will remind you of the method.

WORKED EXAMPLE 1

Find the empirical formula of the compound that has the following composition by mass:

C 40.00% H 6.67% O 53.33%

Answer

C (40.00 ÷ 12.0) = 3.33

H (6.67 ÷ 1.0) = 6.67

O (53.33 ÷ 16.0) = 3.33

Dividing all three numbers by the smallest (i.e. 3.33) gives the following ratio:

C : H : O is 1 : 2 : 1

The empirical formula = CH_2O.

If we know the molecular mass (or molar mass in g mol^{-1}) of the compound we can find the molecular formula by dividing the molecular mass by the empirical formula mass. The molecular formula is always a simple multiple of the empirical formula.

Let us assume that this compound has a molecular mass of 60.0.

The empirical formula mass = (12.0 + 2.0 + 16.0) = 30.0.

(60.0 ÷ 30.0) = 2

The molecular formula must therefore contain twice as many atoms of each element.

The molecular formula = $C_2H_4O_2$.

COMBUSTION ANALYSIS

We can use combustion analysis to find the percentage composition of a compound.

A known mass of the compound is burned in excess dry oxygen. The masses of carbon dioxide and water produced are measured.

For a compound that contains carbon, hydrogen and oxygen only we can then work out the following:

- mass of carbon dioxide → mass of carbon → % carbon
- mass of water → mass of hydrogen → % hydrogen
- % oxygen = 100 – (% carbon + % hydrogen).

WORKED EXAMPLE 2

2.90 g of an organic compound, containing carbon, hydrogen and oxygen only, was burned in excess oxygen.

6.60 g of carbon dioxide and 2.70 g of water were produced.

Calculate the percentage composition of the compound.

Answer

6.60 g of carbon dioxide contains $\frac{12.0}{44.0} \times 6.60 = 1.80$ g of carbon.

$$\% \text{ carbon} = \frac{1.80}{2.90} \times 100 = 62.1\%$$

2.70 g of water contains $\frac{2.0}{18.0} \times 2.70 = 0.30$ g of hydrogen.

$$\% \text{ hydrogen} = \frac{0.30}{2.90} \times 100 = 10.3\%$$

% oxygen = 100 – (62.1 + 10.3) = 27.6%

The percentage composition by mass of the compound is:

C 62.1% H 10.3% O 27.6%

CHARACTERISTIC REACTIONS OF FUNCTIONAL GROUPS

To determine the structural formula from the molecular formula we need to identify the functional groups in the compound. Traditionally, this is done by performing reactions in test tubes, using test reagents such as bromine water or 2,4-dinitrophenylhydrazine.

You have already seen details of these tests in **Book 1** and in **Topics 15**, **18** and **19**.

A reminder of these tests is given in **table A**.

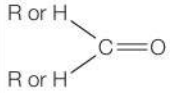
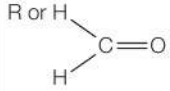
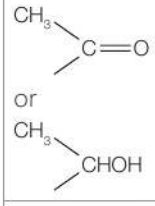
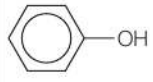
FUNCTIONAL GROUP	TEST	OBSERVATION	NOTES
Alkene (C=C)	add $Br_2(aq)$	bromine decolourises immediately	aldehydes and ketones decolourise $Br_2(aq)$, but very slowly phenols decolourise $Br_2(aq)$ immediately and also produce a white precipitate
Halogenoalkanes (R–X), where X = Cl, Br or I	warm with NaOH(aq) and then add dilute HNO_3, followed by a few drops of $AgNO_3(aq)$	R–Cl produces a white precipitate R–Br produces a cream precipitate R–I produces a yellow precipitate	the precipitate is insoluble in dilute $NH_3(aq)$ the precipitate is insoluble in dilute $NH_3(aq)$, but soluble in concentrated $NH_3(aq)$ the precipitate is insoluble in both dilute $NH_3(aq)$ and concentrated $NH_3(aq)$
Hydroxy group (–OH)	add solid PCl_5	misty fumes	both alcohols and carboxylic acids, but not phenols, produce misty fumes
Primary alcohol (RCH_2OH)	add acidified $K_2Cr_2O_7$ solution and warm	solution turns from orange to green	the organic product of this reaction produces a silver mirror with Tollens' reagent
Secondary alcohol (R_2CHOH)	add acidified $K_2Cr_2O_7$ solution and warm	solution turns from orange to green	the organic product does not produce a silver mirror with Tollens' reagent
Carbonyl group 	add 2,4–dinitrophenylhydrazine	orange precipitate	both aldehydes and ketones produce an orange precipitate
Aldehydes 	add Tollens' reagent and warm	silver mirror forms	Fehling's or Benedict's solutions can also be used ketones do not react with Tollens' reagent, Fehling's solution or Benedict's solution
	add an alkaline solution of iodine and warm	yellow precipitate forms	this is known as the triiodomethane (or iodoform) test
Carboxylic acids (R–COOH)	add $NaHCO_3(aq)$ or $Na_2CO_3(aq)$ and warm if necessary	bubbles of gas	
Phenol 	add bromine water	bromine immediately decolourises and a white precipitate forms	phenylamine produces the same observation however, phenol is soluble in NaOH(aq) but insoluble in dilute HCl; phenylamine is insoluble in NaOH(aq) but soluble in dilute HCl

table A Tests for functional groups.

MODERN ANALYTIC TECHNIQUES

Chemists use a combination of techniques to identify organic compounds and determine their structures, including:

- mass spectra
- infrared spectra
- NMR spectra (both ^{13}C and proton).

Mass spectrometry allows us to determine the accurate relative molecular mass of a compound. The fragmentation patterns also allow us to suggest possible groups that may be present in the compound. For example, a peak with an m/z value of 15 could indicate that a CH_3 group is present.

Infrared spectroscopy allows us to identify particular functional groups and features in an organic compound. For example, an absorption between 1740 and 1720 cm^{-1} indicates the presence of a C=O group in an aldehyde.

^{13}C NMR spectroscopy allows us identify the number of different environments for the carbon atoms in a compound. For example, the spectrum for ethanol, CH_3CH_2OH, has two peaks. This shows that there are two different environments for the carbon atoms.

The ^{13}C NMR spectrum of 1-methylethyl propanoate, $CH_3CH_2COOCH(CH_3)_2$, has five peaks. This shows that there are five different environments for the carbon atoms. If you are wondering why there are only five peaks but six carbon atoms, it is because the carbons in the two methyl groups at the right-hand end of the molecule are in identical environments.

Proton NMR spectroscopy allows us to identify groups of hydrogen atoms in a compound. For example, if there are three hydrogen atoms in the same environment, this indicates that a $-CH_3$ group is present. Splitting patterns also allow us to identify the number of hydrogen atoms on an adjacent carbon atom.

None of these methods gives a complete answer, but together they provide good evidence to allow you to identify the molecular structure.

Mass spectrometry and infrared spectroscopy were introduced in **Topic 10 (Book 1: IAS)**. NMR spectroscopy was introduced in **Topic 15**.

It may be useful to re-read these two topics so that you fully understand the two worked examples.

WORKED EXAMPLE 3

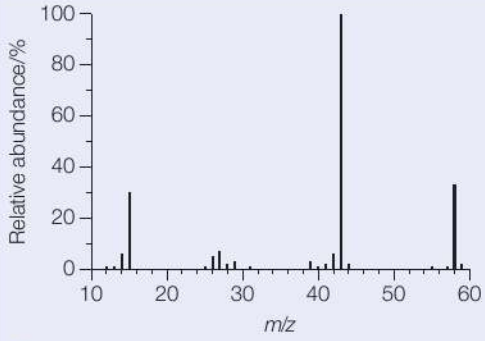
Compound X has the following percentage composition by mass:

C 62.07% H 10.34% O 27.59%

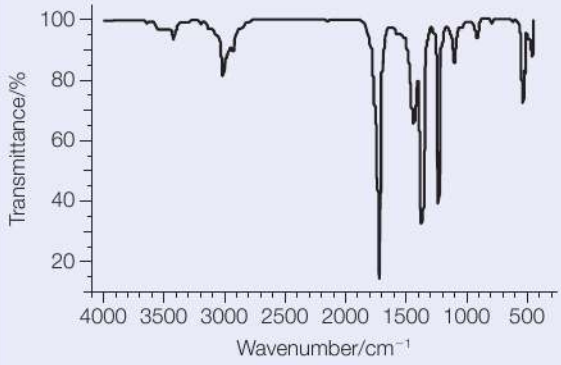
Compound X produces an orange precipitate with 2,4-dinitrophenylhydrazine, but it does not react with either Fehling's solution or Tollens' reagent.

Fig A shows the mass spectrum of compound X.

Fig B shows the infrared spectrum of compound X.



▲ **fig A** Mass spectrum of compound X.



▲ **fig B** Infrared spectrum of compound X.

The proton NMR spectrum of compound X has a single peak with a chemical shift of 2.2 ppm. The peak area corresponds to six carbon atoms.

Use all this information to determine the structure of compound X.

Answer

Calculation of the empirical formula:

	Divide by A_r	*Divide by smallest*
C	(62.07 ÷ 12.0) = 5.17	(5.17 ÷ 1.72) = 3
H	(10.34 ÷ 1.0) = 10.34	(10.34 ÷ 1.72) = 6
O	(27.59 ÷ 16.0) = 1.72	(1.72 ÷ 1.72) = 1

The empirical formula is C_3H_6O.

The molecular ion peak in the mass spectrum has a value of 58. The relative molecular mass of compound X is therefore 58.

The empirical formula mass is (12.0 x 3) + (1.0 x 6) + 16.0 = 58.0.

Therefore the molecular formula of compound X is C_3H_6O.

The absorption at 1720 cm^{-1} indicates that a C=O group is present. This could be in either an aldehyde or a ketone.

Compound X produces an orange precipitate with 2,4-dinitrophenylhydrazine, showing that it is a carbonyl compound (i.e. it is either an aldehyde or a ketone).

However, it does not react with either Fehling's solution or Tollens' reagent, showing that it must be a ketone.

The proton NMR spectrum shows that there are six hydrogen atoms in identical environments, indicating that there are two CH_3 groups attached to the third carbon atom. The chemical shift of 2.2 ppm indicates hydrogen atoms in the environment: H–C–C=O.

So the structure of compound X is CH_3COCH_3 (propanone).

WORKED EXAMPLE 4

(a) An organic compound **Y** contains carbon, hydrogen, nitrogen and oxygen.

0.132 g of **Y** is burned completely in oxygen to produce 0.176 g of carbon dioxide, 0.072 g of water and 24.0 cm^3 of nitrogen (molar volume of nitrogen under the conditions of the experiment = 24.0 $dm^3\ mol^{-1}$).

The relative molecular mass of compound **Y** is 132.

Determine the molecular formula of **Y**.

(b) A sample of **Y** is hydrolysed by heating under reflux with concentrated hydrochloric acid for several hours. The reaction mixture is then cooled and neutralised. The only organic product, **Z**, of this reaction has the molecular formula $C_2H_5O_2N$.

One mole of **Z** reacts with either one mole of hydrochloric acid or one mole of sodium hydroxide.

Z can exist as a zwitterion.

Use all the information above to deduce the structures of compounds **Z** and **Y**.

Answer

(a) Step 1: *calculate mass of oxygen*

$$\text{mass of carbon} = \frac{12.0}{44.0} \times 0.176 = 0.048\ \text{g}$$

$$\text{mass of hydrogen} = \frac{2.0}{18.0} \times 0.072 = 0.008\ \text{g}$$

mass of nitrogen = (24.0 ÷ 24 000) × 28.0 = 0.028 g

mass of oxygen = 0.132 g − (0.008 + 0.048 + 0.028) g
= 0.048 g

Step 2: *calculate empirical formula of* **Y**

moles of carbon = (0.048 ÷ 12.0) = 0.004

moles of hydrogen = (0.008 ÷ 1.0) = 0.008

moles of nitrogen = (0.028 ÷ 14.0) = 0.002

moles of oxygen = (0.048 ÷ 16.0) = 0.003

Empirical formula of **Y** is $C_4H_8N_2O_3$.

Empirical formula mass
= (12.0 x 4) + (1.0 x 8) + (14.0 x 2) + (16.0 x 3)
= 132

The empirical formula mass = the relative molecular mass

Therefore, the molecular formula of Y is $C_4H_8N_2O_3$.

(b) **Z** reacts with HCl, so it contains an amine group ($-NH_2$).

Z reacts with NaOH, so it contains a carboxylic acid group (−COOH).

There is one $-NH_2$ group per molecule and one −COOH per molecule (one mole of **Z** reacts with both one mole of HCl and one mole of NaOH).

Z can form a zwitterion, so it must be an amino acid.

The structure of **Z** is H_2NCH_2COOH.

Z is produced by hydrolysis of **Y**. **Y** must therefore be an amide.

Since there is only one organic product, the structure of **Y** must be $H_2NCH_2CONHCH_2COOH$.

CHECKPOINT

SKILLS ANALYSIS

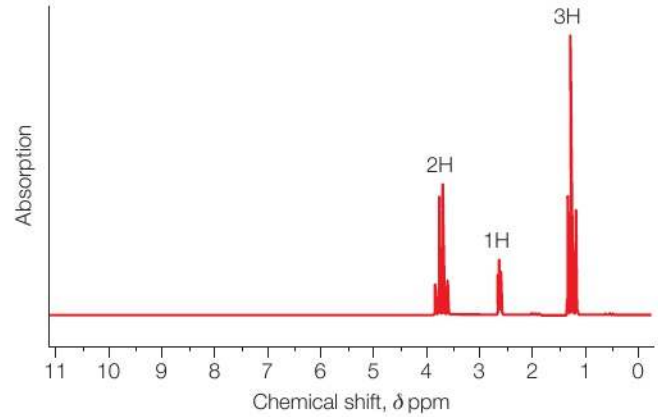
1. Compound **T** does not immediately decolourise bromine water. It also does not react with an aqueous solution of sodium carbonate. When excess PCl_5 is added it produces one mole (1 mol) of hydrogen chloride gas per mole of **T**. It produces an orange precipitate when added to 2,4-dinitrophenylhydrazine.

T has a molecular formula of $C_3H_6O_2$.

(a) Write the structural formulae of three possible isomers of **T**. Justify your answer.

(b) The mass spectrum of **T** has peaks at *m/z* values of 29 and 59. Use this information to deduce the identity of **T**.

2. The proton NMR spectrum of an organic compound is shown below.



The percentage composition by mass of the organic compound is:

C 52.17% H 13.04% O 34.78%

It has a relative molecular mass of 46.

Use this information to identify the compound.

20A 2 ORGANIC SYNTHESIS

SPECIFICATION REFERENCE
20.2 20.3(i) 20.3(ii)

LEARNING OBJECTIVES

- Understand methods of increasing the length of the carbon chain in a molecule by the use of magnesium to form Grignard reagents and the reactions of the latter with carbon dioxide and with carbonyl compounds in dry ether.
- Be able to use knowledge of organic chemistry to solve problems such as:
  (i) predicting the properties of unfamiliar compounds containing one or more of the functional groups included in the specification and explain these predictions
  (ii) planning reaction schemes of up to four steps, recalling familiar reactions and using unfamiliar reactions given sufficient information.

WHAT IS ORGANIC SYNTHESIS?

The term 'synthesis' (the plural is 'syntheses') refers to making something new from what already exists. In organic chemistry it usually means using a familiar compound to make an unfamiliar compound. You have already learned many examples of converting one compound into another – for example, converting a halogenoalkane into an alcohol. In this section we will consider converting one compound into another compound that cannot be achieved in one step, but needs two or more steps. To keep things relatively simple, in this book the number of steps will be limited to four.

The ability to plan a reaction scheme to achieve this synthesis is not easily learned. It is not like recalling a simple mathematical expression and putting the numbers into a calculator to obtain the answer. Instead, it relies on a thorough knowledge of a range of chemical reactions and the ability to select the ones that are appropriate to the problem.

Here is a simple example of a two-step synthesis – how to convert bromoethane to ethyl propanoate. You could start by considering the reactions of bromoethane, and then consider reactions in which ethyl propanoate is a product. You could consider the question as a puzzle in which you need to find the identity of X in this sequence:

bromoethane → X → ethyl propanoate

If you know a product of a reaction of bromoethane that is the same as a compound that can be converted into ethyl propanoate, then you only need to include the reagents and conditions to achieve each of these two steps. When the synthesis involves three steps it is more difficult because you need to identify two intermediate compounds in the synthesis. If there are four steps, then we have reached the limit of what is reasonable to expect at this point in your study of chemistry!

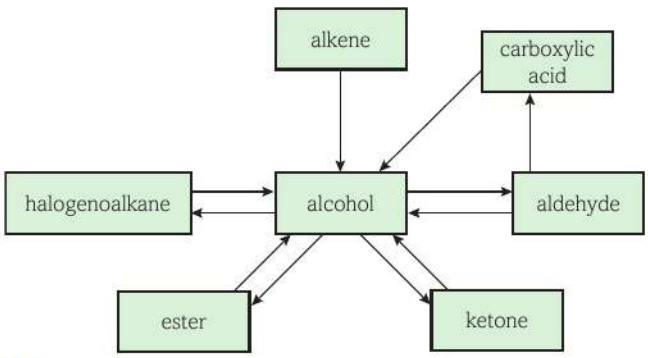
In the example above, X is ethanol, and the steps are:

Step 1: bromoethane → ethanol
reflux with aqueous potassium hydroxide (hydrolysis)

Step 2: ethanol → ethyl propanoate
heat with propanoic acid using an acid catalyst (esterification).

PREPARING FOR PLANNING A SYNTHESIS

One thing that will help you learn how to plan a synthesis is to put together an outline of all the reactions you need to know. You could leave out reactions that are less useful in synthesis, such as combustion, polymerisation and chemical tests. You could try to produce a diagram (like a spider diagram) that would include all the reactions, but that would probably be too complicated to easily use. It might be better to produce a series of smaller ones. For example, **fig A** shows reactions based on alcohols.



▲ **fig A** Reactions of alcohols.

Where the arrows point in both directions, this means that you need to know both conversions. Note that reactions from **Book 1** are included. If you copy this onto a sheet of paper you can add the formulae of reagents and any special conditions on the arrows. You could use different colours to emphasise different features.

EXTENDING A CARBON CHAIN

It is possible that one of the steps in your synthesis will involve extending an existing carbon chain by one or more carbon atoms. You have already learned three ways of doing this:

1 Reacting a halogenoalkane with a cyanide ion forms a nitrile with one more carbon atom than the halogenoalkane.
2 The addition of hydrogen cyanide to a carbonyl compound.
3 The alkylation of benzene, which introduces an alkyl group into a benzene ring.

A different way, which is more versatile, involves the use of Grignard reagents.

GRIGNARD REAGENTS

Victor Grignard was a French chemist who devised a new way to synthesise organic compounds. His method involved the use of organometallic compounds containing magnesium, which are now known as Grignard reagents. They are made by heating under reflux the chosen halogenoalkane with magnesium in a solvent of dry ether. Halogenoarenes such as bromobenzene are also used. Bromoalkanes are used in preference to other halogenoalkanes. Grignard reagents contain magnesium covalently bonded to both the alkyl group and the halogen. A general equation for the formation of a Grignard reagent is:

$$R{-}Br + Mg \rightarrow R{-}Mg{-}Br$$

Grignard reagents react with water, so they are both made and used in a solvent of dry ether.



▲ **fig B** Victor Grignard (1871–1935) won the Nobel Prize in Chemistry for his work in organic synthesis.

GRIGNARD REACTIONS

Once a Grignard reagent has been made it can then be converted into a range of organic compounds, depending on the reagent used. In this book we will consider only two of these reactions: with carbon dioxide and with carbonyl compounds. They can be summarised as follows.

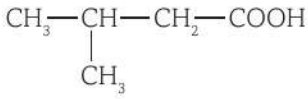
1	with carbon dioxide	$RMgBr \rightarrow RCOOH$	carboxylic acid
2a	with methanal	$RMgBr \rightarrow RCH_2OH$	primary alcohol
2b	with an aldehyde R′CHO	$RMgBr \rightarrow RR'CHOH$	secondary alcohol
2c	with a ketone R′COR″	$RMgBr \rightarrow RR'R''COH$	tertiary alcohol

R, R′ and R″ represent different alkyl groups, although they could be the same. Note that both carbon dioxide and methanal increase the carbon chain by one carbon atom. Both aldehydes and ketones produce a branch, of variable length, on the existing carbon chain.

Having made the appropriate Grignard reagent, the second reagent (carbon dioxide or the carbonyl compound) is added. After the reaction is complete, a dilute acid is added to obtain the desired organic product.

EXAMPLES OF GRIGNARD REACTIONS

FORMATION OF 3-METHYLBUTANOIC ACID



This is an example of reaction 1, described above. The COOH part comes from carbon dioxide, so the starting compound must supply the $(CH_3)_2CHCH_2$ part, which means using $(CH_3)_2CHCH_2Br$, i.e. 1-bromo-2-methylpropane. The equations for the reactions that occur in this example are:

Step 1: formation of the Grignard reagent

$$(CH_3)_2CHCH_2Br + Mg \rightarrow (CH_3)_2CHCH_2MgBr$$

Step 2: reaction with chosen reagent

$$(CH_3)_2CHCH_2MgBr + CO_2 \rightarrow (CH_3)_2CHCH_2COOMgBr$$

Step 3: hydrolysis using a dilute acid

$$(CH_3)_2CHCH_2COOMgBr + H_2O \rightarrow (CH_3)_2CHCH_2COOH + Mg(OH)Br$$

The inorganic product of Step 3 will react with the dilute acid. You do not need to know details of the mechanisms.

FORMATION OF PROPAN-1-OL

This and the other examples are abbreviated as reaction schemes, rather than being shown as three steps.

Propan-1-ol, $CH_3CH_2CH_2OH$, is a primary alcohol. So, methanal is the source of the CH_2OH part, and the starting compound is bromoethane, which is the source of the CH_3CH_2 part.

$$CH_3CH_2Br \xrightarrow{Mg/ether} CH_3CH_2MgBr \xrightarrow[\text{then } H_2O/H^+]{HCHO} CH_3CH_2CH_2OH$$

EXAM HINT

Grignard reactions are a lot safer than using hydrogen cyanide and potassium cyanide to add a carbon atom to a molecule (see **Section 20A.3**). Grignard reactions also give a wider range of chemical additions.

EXAM HINT

You can test your knowledge of organic synthesis and analysis by setting yourself a synthesis task. For example, think about how to convert a three-carbon alkene into a five-carbon secondary alcohol. Then try to **predict** the changes in the NMR spectra from reactant to product.

LEARNING TIP

Practise choosing reagents for a Grignard synthesis. First, identifying the product type (acid or alcohol) tells you the type of compound to react with the Grignard reagent. Then, work out how many carbon atoms are needed in the halogenoalkane used to form the Grignard reagent. See how you get on with the questions in this section!

EXAM HINT

You have seen in **Topic 10 (Book 1: IAS)** that primary alcohols are easily oxidised by heating with a mixture of potassium dichromate(VI) and dilute sulfuric acid. There is a colour change from orange to green in this reaction.

You may have learned that phenols are not easily oxidised. So you may think that 4-methylphenol will not be oxidised by heating with the same mixture of potassium dichromate(VI) and dilute sulfuric acid. You may decide to use this reagent to distinguish between the two compounds.

Unfortunately, this test will not work. Although the phenol group will not be oxidised by the mixture of potassium dichromate(VI) and dilute sulfuric acid, the methyl group attached to the benzene ring *will* be oxidised. So a colour change from orange to green will be observed.

This is a good warning to be very careful when choosing reagents for distinguishing between two compounds. The best advice is to use only the reactions described in **Book 1** and **Book 2**.

FORMATION OF PENTAN-2-OL

Pentan-2-ol, $CH_3CH(OH)CH_2CH_2CH_3$, is a secondary alcohol, so ethanal is the source of the $CH_3CH(OH)$ part, and the starting compound is 1-bromopropane, which is the source of the $CH_3CH_2CH_2$ part.

$$CH_3CH_2CH_2Br \xrightarrow{Mg/ether} CH_3CH_2CH_2MgBr \xrightarrow[\text{then } H_2O/H^+]{CH_3CHO} CH_3CH(OH)CH_2CH_2CH_3$$

FORMATION OF 2-METHYLPROPAN-2-OL

2-Methylpropan-2-ol, $(CH_3)_3COH$, is a tertiary alcohol, so propanone is the source of the $(CH_3)_2COH$ part, and the starting compound is bromomethane, which is the source of the CH_3 part.

$$CH_3Br \xrightarrow{Mg/ether} CH_3MgBr \xrightarrow[\text{then } H_2O/H^+]{CH_3COCH_3} (CH_3)_3COH$$

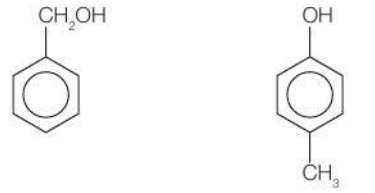
PREDICTING THE CHEMICAL PROPERTIES OF COMPOUNDS

As we have seen in **Section 20A.1**, chemical reagents can be used to show which functional groups are present in a compound.

Functional groups can influence the behaviour of compounds that contain them. For example:

- some compounds will act as nucleophiles and others as electrophiles
- some will be susceptible to addition reactions and others to substitution reactions
- some will be easily oxidised and others are more easily reduced.

We can show this by looking at two compounds with the molecular formula C_7H_8O: phenylmethanol and 4-methylphenol (see **fig C**).



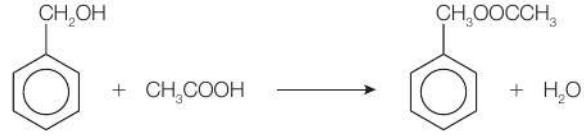
phenylmethanol 4-methylphenol

▲ **fig C**

Both compounds are slightly soluble in water. Phenylmethanol forms a neutral solution, but aqueous 4-methylphenol is slightly acidic. This is because one of the lone pairs of electrons on the oxygen atom is incorporated into the delocalised electrons in the ring. This makes the O–H bond more polar (see **Topic 18**). However, 4-methylphenol is not acidic enough to react with sodium hydrogen carbonate to form carbon dioxide, distinguishing it from carboxylic acids.

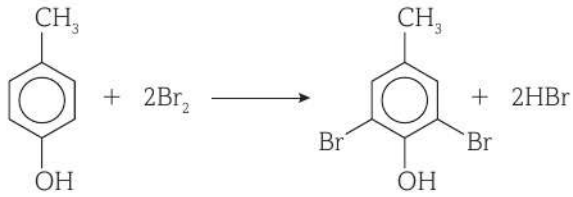
Phenylmethanol is a primary alcohol because it contains a $-CH_2OH$ group. 4-Methylphenol is a phenol because it has an –OH group attached to the benzene ring.

Phenylmethanol will form an ester when heated with a carboxylic acid in the presence of concentrated sulfuric acid. This is because it is an alcohol (**Topic 15**):



4-Methylphenol will not form an ester with carboxylic acids.

4-Methylphenol will decolourise bromine water:



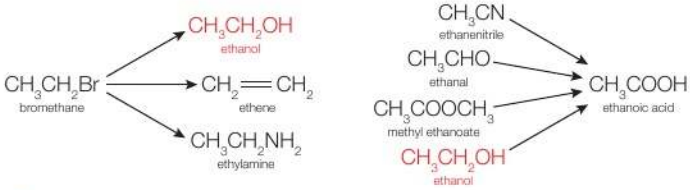
However, phenylmethanol does not react with bromine water.

MULTI-STEP SYNTHESES

You may be asked to plan to a reaction scheme that shows all the intermediate compounds as well as the reagents and conditions required for each step. There are several ways you can approach this:

- by checking if the carbon chain length has increased or decreased
- by looking at the final product and 'working back' to the starting compound
- by looking at the starting compound and thinking of the types of reactions it can undergo.

Fig D shows how working back can be used to plan the synthesis of one compound from another. In this case the starting compound is bromoethane and the final compound is ethanoic acid:



▲ **fig D**

You can plan the reaction scheme using the following stages.

- Begin by writing the formulae of the compounds that can be readily converted into ethanoic acid. These include ethanenitrile, ethanol, methyl ethanoate and ethanol.
- Now look at the starting compound, bromoethane, to see if it can be converted into one of these compounds. If necessary, write the formulae of the compounds that bromoethane can easily be converted into. These include ethanol, ethene and ethylamine.
- You should now be able to see a clear two-step route from bromoethane to ethanoic acid. In this case, the route is to convert bromoethane into ethanol, and then to convert ethanol into ethanoic acid.
- Finally, you need to add the reagents and conditions required for each step. To convert bromoethane into ethanol you need to add aqueous sodium hydroxide and heat under reflux. To convert ethanol into ethanoic acid you need to add potassium dichromate(VI) and dilute sulfuric acid and heat under reflux.

You can set out your final answer as shown in **fig E**:

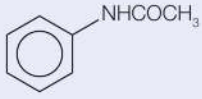


▲ **fig E**

You can use the same process for three- and four-step syntheses, although it will be a bit more complex.

WORKED EXAMPLE 1

Deduce the reagents and conditions, and identify the two intermediates, in the three-step synthesis of the following compound from benzene:



Answer

Working backwards

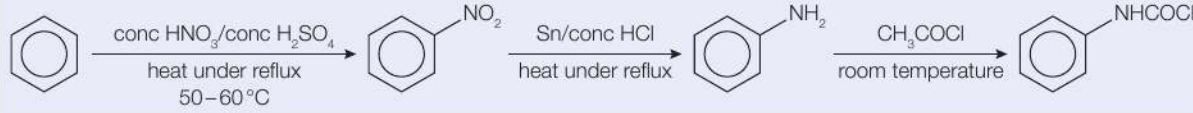
The final compound is an *N*-substituted amide (a secondary amide). It can be made by the reaction between an amine and an acyl chloride, at room temperature. In this case, the reactants are phenylamine, $C_6H_5NH_2$, and ethanoyl chloride, CH_3COCl.

Phenylamine can be made by heating nitrobenzene, $C_6H_5NO_2$, under reflux with tin and concentrated hydrochloric acid.

Working forwards

Benzene, C_6H_6, is converted into nitrobenzene by heating under reflux at 50–60 °C with a mixture of concentrated nitric acid and concentrated sulfuric acid.

The three-step synthesis is therefore:



WORKED EXAMPLE 2

Deduce the reagents and conditions, and identify the three intermediates, in the four-step synthesis of ethanol, CH_3CH_2OH, from methanol, CH_3OH.

Answer

The first point to note is the increase in carbon chain length from 2 to 3. You have seen three methods for increasing the carbon chain:

- reaction between a halogenoalkane and KCN
- reaction between an alkene and HCN
- the use of Grignard reagents.

We are going to look at how to do this conversion without the use of a Grignard reagent.

Working backwards

There are several ways of making ethanol. For example:

- hydrolysing chloroethane, CH_3CH_2Cl, or bromoethane, CH_3CH_2Br
- reducing ethanal, CH_3CHO
- reducing ethanoic acid, CH_3COOH.

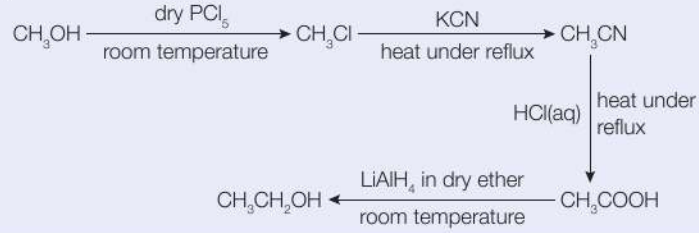
Ethanoic acid can be made by hydrolysing ethanenitrile, CH_3CN.

Converting a halogenoalkane into a nitrile is one method of increasing the carbon chain. CH_3CN can be made by reacting CH_3Cl with KCN.

Working forwards

CH_3OH can be converted into CH_3Cl by reaction with PCl_5.

A possible four-step synthesis is therefore:



CHECKPOINT

SKILLS DECISION-MAKING, REASONING

1. Draw the structure of the organic product obtained in a Grignard synthesis starting from 1-bromo-2,2-dimethylpropane and propanal.

2. Draw the structures of the two compounds needed in a Grignard synthesis of 3,3-dimethylhexan-4-ol.

3. Deduce the reagents and conditions, and identify all intermediates, in each of the following conversions:

(a) CH_3Br to $CH_3CH_2NH_2$ (two steps)

(b) $CH_2{=}CH_2$ to $CH_3CH_2CH_2OH$ (four steps).

20A 3 HAZARDS, RISKS AND CONTROL MEASURES

SPECIFICATION REFERENCE
20.3(iii) 20.3(iv)

LEARNING OBJECTIVES

- Be able to use knowledge of organic chemistry to solve problems such as:
  (i) selecting suitable practical procedures for carrying out reactions involving compounds with functional groups
  (ii) identifying appropriate control measures to reduce risk based on data for hazards.

CONVERTING ONE FUNCTIONAL GROUP INTO ANOTHER

This is a summary of some of the most useful organic conversions that you have learned on your International AS/A Level course.

CONVERSION	EQUATION	REAGENT	CONDITIONS
Alkene to halogenoalkane	$CH_2{=}CH_2 + HX \rightarrow CH_3CH_2X$	hydrogen halide	mix the gases at room temperature
Halogenoalkane to alcohol	$RX + NaOH \rightarrow ROH + NaX$	aqueous sodium hydroxide	heat under reflux
Halogenoalkane to nitrile	$RX + KCN \rightarrow RCN + KX$	alcoholic potassium cyanide	heat under reflux
Halogenoalkane to amine	$RX + 2NH_3 \rightarrow RNH_2 + NH_4X$	aqueous ammonia	heat under pressure
Alcohol to chloroalkane	$ROH + PCl_5 \rightarrow RCl + HCl + POCl_3$	phosphorus(V) chloride	room temperature
Alcohol to bromoalkane	$ROH + HBr \rightarrow RBr + H_2O$	50% concentrated sulfuric acid and potassium bromide	warm
Alcohol to iodoalkane	$3ROH + PI_3 \rightarrow 3RI + H_3PO_3$	red phosphorus and iodine	heat under reflux
Primary alcohol to aldehyde	$RCH_2OH + [O] \rightarrow RCHO + H_2O$	potassium dichromate(VI) and dilute sulfuric acid	add the reagent to hot alcohol and allow the aldehyde to distil off as it is formed
Primary alcohol to carboxylic acid	$RCH_2OH + 2[O] \rightarrow RCOOH + H_2O$	potassium dichromate(VI) and dilute sulfuric acid	heat under reflux
Secondary alcohol to ketone	$RCH(OH)R' + [O] \rightarrow RCOR' + H_2O$	potassium dichromate(VI) and dilute sulfuric acid	heat under reflux
Aldehyde to primary alcohol	$RCHO + 2[H] \rightarrow RCH_2OH$	lithium aluminium hydride in dry ether	room temperature
Ketone to secondary alcohol	$RCOR' + 2[H] \rightarrow RCH(OH)R'$	lithium aluminium hydride in dry ether	room temperature
Aldehyde and ketone to 2-hydroxynitrile	$RCHO + HCN \rightarrow RCH(OH)CN$ $RCOR' + HCN \rightarrow RR'C(OH)CN$	potassium cyanide in dilute sulfuric acid	10–20 °C
Carboxylic acid to primary alcohol	$RCOOH + 4[H] \rightarrow RCH_2OH + H_2O$	lithium aluminium hydride in dry ether	room temperature
Carboxylic acid to ester	$RCOOH + R'OH \rightarrow RCOOR' + H_2O$	alcohol and concentrated sulfuric acid	heat
Acyl chloride to carboxylic acid	$RCOCl + H_2O \rightarrow RCOOH + HCl$	water	room temperature
Acyl chloride to ester	$RCOCl + R'OH \rightarrow RCOOR' + HCl$	alcohol	room temperature
Acyl chloride to primary amide	$RCOCl + 2NH_3 \rightarrow RCONH_2 + NH_4Cl$	aqueous ammonia	room temperature
Acyl chloride to secondary amide (*N*-substituted amide)	$RCOCl + R'NH_2 \rightarrow RCONHR' + HCl$	amine	room temperature

CONVERSION	EQUATION	REAGENT	CONDITIONS
Nitrile to primary amine	$RCN + 4[H] \rightarrow RCH_2NH_2$	lithium aluminium hydride in dry ether	room temperature
Nitration of benzene	$C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O$	concentrated nitric acid and concentrated sulfuric acid	heat under reflux between 50 and 60 °C
Sulfonation of benzene	$C_6H_6 + H_2SO_4 \rightarrow C_6H_5SO_3H + H_2O$	fuming sulfuric acid	40 °C
Bromination of benzene	$C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr$	liquid bromine with iron (to form iron(III) bromide)	dry and room temperature
Friedel–Crafts alkylation of benzene	$C_6H_6 + RX \rightarrow C_6H_5R + HX$	halogenoalkane	dry, in the presence of anhydrous aluminium halide
Friedel–Crafts acylation of benzene	$C_6H_6 + RCOCl \rightarrow C_6H_5COR + HCl$	acyl chloride	dry, in the presence of anhydrous aluminium chloride

SAFETY IN CHEMISTRY LABORATORIES

Incidents that cause harm to people are rare in school and college chemistry laboratories. One of the reasons for this is that all laboratories must consider the hazards of carrying out chemistry experiments and use safe methods of working.

This applies to all chemistry experiments, whether you are given the instructions for doing them, or whether you plan your own. When you plan an organic synthesis you must consider the hazards associated with the reactants, the substance you are synthesising and with any intermediate compounds formed.

HAZARDS, RISKS AND CONTROL MEASURES

The hazards of chemical substances relate to the inherent properties of each substance and the way in which it will be used. Most people would consider that water (H_2O) is completely safe and has no hazards. In most situations this is the case. However, consider a beaker of boiling water. The steam coming from the boiling water, and the boiling water itself, could both cause harmful effects if they came into contact with your skin.

Now consider a substance that most people would consider dangerous. You probably used dilute hydrochloric acid years ago in experiments involving marble chips. You will have been told to wear eye protection when using the acid, because of the harm it could do if it got into your eyes.

The **hazard** exists because hydrochloric acid is corrosive.

The **risk** is that hydrochloric acid may get into your eyes and cause harm.

The control measure is to wear eye protection to reduce the risk of the acid getting into your eyes.

This simple example should help you to understand the difference between hazard and risk – these two words are often thought, wrongly, to have the same meaning.

HAZARD WARNING SYMBOLS

A long time ago, bottles containing certain chemicals were labelled with the word POISON. This early attempt to prevent harm to laboratory workers was well intentioned, but some people may have thought that bottles without such a label contained harmless substances.

More recently, symbols (sometimes called pictograms) have been used as labels for bottles to identify the actual hazard of the substance inside. The actual symbols have changed over the years, and you may still see older ones used, especially those in squares with an orange background, such as these:



The symbols in current use are red diamond shapes. **Table A** shows some of the more common ones, along with a simple description of their meanings.

SYMBOL	MEANING	
	Health hazard	includes warnings on skin rashes, eye damage and ingestion
	Corrosive	can cause skin burns and permanent eye damage
	Flammable	can catch fire if heated or comes into contact with a flame
	Acute toxicity	can cause life-threatening effects, even in small quantities

table A

In some cases, the substance may
especially when it is an aqueous so
likely to use hydrochloric acid in th

- when used in a titration, it will h approximately 0.1 mol dm^{-3}
- as a general laboratory reagent, it will probably have a concentration of approximately 1 or 2 mol dm^{-3}
- a bottle labelled 'concentrated hydrochloric acid' may have a concentration of more than 10 mol dm^{-3}.

LEARNING TIP

Become familiar with the hazards of the substances you use in an experiment, and consider how any risks of using them may be reduced.

These very different concentrations have different risks.

There are several other hazard symbols, including those that you are more likely to find in biology or physics laboratories.

CONTROL MEASURES

It is the responsibility of whoever is in charge of the laboratory to identify risks and hazards, and to prescribe appropriate control measures. This responsibility is shared with the student doing the experiment, who should follow the guidance. There are several organisations that provide advice and support to schools and colleges, and it is likely that the health and safety information you see in your laboratory has come from one of these sources.

As a student you may be asked to plan an experiment or a synthesis, including identifying the hazards and control measures needed, so you need to be familiar with this information.



▲ **fig A** You will also find examples of hazards, risks and control measures in your home.

APPARATUS

You also need to consider risks associated with apparatus. For example, mercury thermometers are often replaced by spirit thermometers or digital thermometers, which have much lower risks. Heating may be done using an electrical heating mantle instead of a Bunsen burner. You may use glass apparatus with ground glass joints, as these are safer to set up than those that use corks or bungs. You also have to consider where to support apparatus with a clamp and stand.

CHECKPOINT

SKILLS SELF-REGULATION, RESPONSIBILITY

1. Suggest some control measures you should use in an experiment to convert bromoethane into ethanol.
2. Use your chemical knowledge to suggest why spilling sodium hydroxide solution onto your skin may cause more harm than hydrochloric acid of the same concentration.

SUBJECT VOCABULARY

hazard a property of a substance that could cause harm to a user

risk the possible effect that a substance may cause to a user, and this will depend on factors such as concentration and apparatus. The level of risk is controlled using control measures

20A 4 PRACTICAL TECHNIQUES IN ORGANIC CHEMISTRY: PART 1

SPECIFICATION REFERENCE
20.5(i) 20.5(vi) 20.5(vii)

LEARNING OBJECTIVES

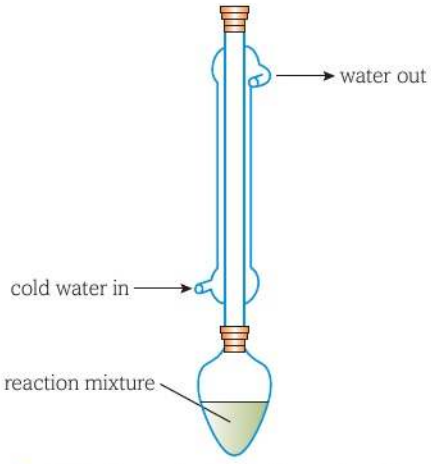
- Understand the following techniques used in the preparation and purification of organic compounds:
  (i) refluxing
  (ii) distillation
  (iii) steam distillation.

DIFFERENT PRACTICAL TECHNIQUES

In the preparation and purification of any organic compound, there are likely to be several techniques used. Each preparation or synthesis will use one or more of the techniques in this section and the next. Most of them have been covered in **Book 1**, but all are included in these sections for completeness.

HEATING UNDER REFLUX

Some reactions involving organic compounds are slow at room temperature, so the obvious thing to do is heat the reaction mixture. However, as many organic compounds are volatile, there is a risk that they will escape from the reaction mixture during the heating process. The normal way to prevent this happening is to heat the reaction mixture in a flask fitted with a reflux condenser (**fig A**). This technique is also known as refluxing. All the vapours rising from the reaction mixture during heating enter the condenser and change back into liquids. They then return to the flask so that the unreacted compounds can react.



▲ **fig A** Apparatus for heating under reflux.

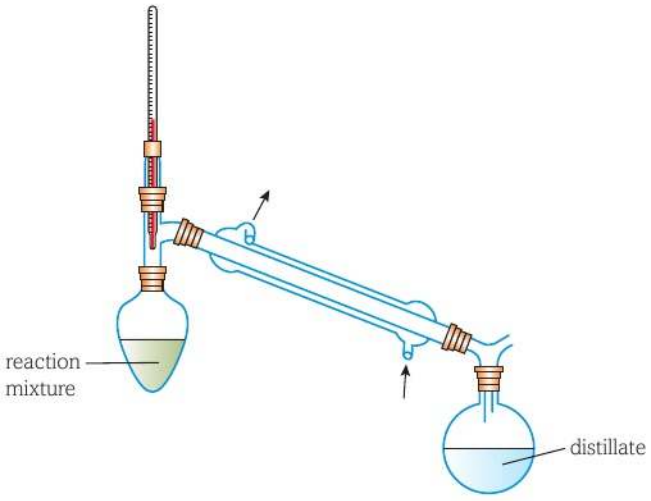
The flask can be heated using hot water or oil in a beaker heated by a Bunsen burner or by using an electric mantle. The use of anti-bumping granules in the reaction mixture helps to make the boiling smooth. A gentle flow of cold water enters at the bottom of the condenser.

METHODS OF SEPARATION

SIMPLE DISTILLATION

This is a separation technique used to obtain a liquid product from a reaction mixture that has a boiling temperature much lower than the other substances in the reaction mixture.

Distillation of an impure liquid involves heating it in a flask connected to a condenser (**fig B**). The liquid with the lowest boiling temperature evaporates or boils off first and passes into the condenser first, and so can be collected in the receiver separately from another liquid that may evaporate later. The purpose of the thermometer is to monitor the temperature of the vapour as it passes into the condenser. If the temperature remains steady, this indicates that one compound is being distilled. If after a while the temperature begins to rise, this indicates that a different compound is being distilled.



▲ **fig B** Apparatus for simple distillation.

The advantages of using simple distillation, rather than fractional distillation, are that it is easier to set up and is quicker. The disadvantage is that it does not separate the liquids as well as fractional distillation. It should only be used if the boiling temperature of the liquid being purified is very different from the other liquids in the mixture, ideally a difference of more than 25 °C.

EXAM HINT

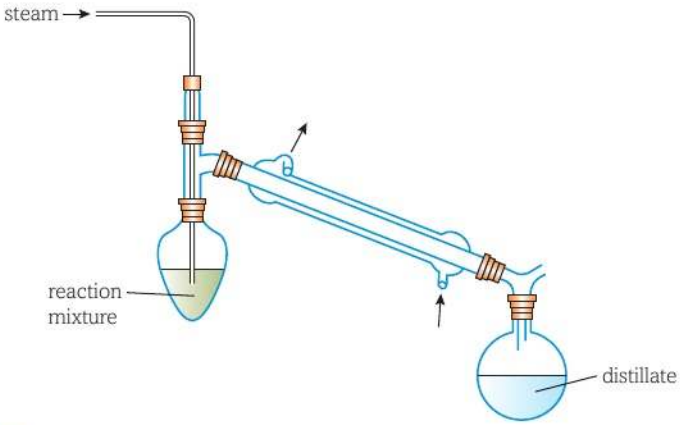
You may be asked to **draw** distillation apparatus (simple or fractional). If so, make sure that you **draw** the bulb of the thermometer level with the side-arm linked to the condenser.

STEAM DISTILLATION

Steam distillation is a technique used to separate an insoluble liquid from an aqueous solution. It involves passing steam into a reaction mixture that contains an aqueous solution and a liquid

that forms a separate layer (**fig C**). The agitation of the liquid caused by the steam bubbling through the mixture ensures that both the insoluble liquid and the aqueous solution are on the surface of the mixture and so can form part of the liquid that evaporates. The advantage of this process is that the insoluble liquid is removed from the reaction mixture at a temperature below its normal boiling temperature.

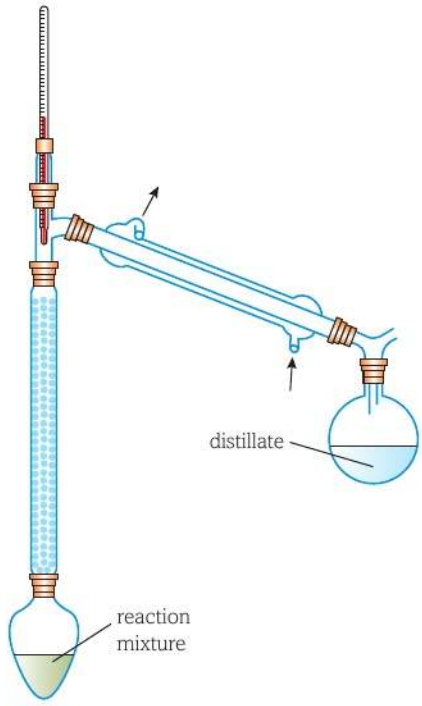
An example is phenylamine, a liquid with a boiling temperature of 184 °C. During steam distillation of a reaction mixture containing phenylamine, a liquid containing phenylamine and water distils at a temperature of 98 °C. An advantage of doing this is that the temperature at which phenylamine distils is much lower than its boiling temperature, so there is less chance of decomposition.



▲ **fig C** Apparatus for steam distillation.

FRACTIONAL DISTILLATION

Fractional distillation uses the same apparatus as simple distillation, but with a fractionating column between the heating flask and the still head (**fig D**). The column is usually filled with glass beads or pieces of broken glass. These act as surfaces on which the vapour leaving the column can condense, and then be evaporated again as more hot vapour passes up the column. Effectively, the vapour experiences several repeated distillations as it passes up the column, which provides a better separation.



▲ **fig D** Apparatus for fractional distillation.

Fractional distillation takes longer than simple distillation. It is best used when the difference in boiling temperatures is small, and when there are several compounds to be separated from a mixture.

LEARNING TIP

Remember that there are three types of distillation, and that you need to consider which one to use in a particular case.

CHECKPOINT

SKILLS SELF-DIRECTION, CREATIVITY

1. Explain briefly why the condenser is vertical in refluxing but nearly horizontal in the three methods of distillation.
2. Suggest, with a reason, which fractionating column provides the better separation – very small glass beads or larger pieces of glass.

20A 5 PRACTICAL TECHNIQUES IN ORGANIC CHEMISTRY: PART 2

SPECIFICATION REFERENCE
20.5(ii) 20.5(iii) 20.5(iv) 20.5(v) 20.5(viii) 20.5(ix)

LEARNING OBJECTIVES

- Understand the following techniques used in the preparation and purification of organic compounds:
  - (i) purification by washing, including with water and sodium carbonate solution
  - (ii) solvent extraction
  - (iii) recrystallisation
  - (iv) drying
  - (v) melting temperature determination
  - (vi) boiling temperature determination.

MORE METHODS OF SEPARATION

SOLVENT EXTRACTION

As the name suggests, this method involves using a solvent to remove the desired organic product from the other substances in the reaction mixture. Several solvents can be used, but the choice depends mainly on these features:

- The solvent added should be immiscible (i.e. does not form a mixture) with the solvent containing the desired organic product.
- The desired organic product should be much more soluble in the added solvent than in the reaction mixture.

The method is this:

- Place the reaction mixture in a separating funnel (**fig A**), and then add the chosen solvent – it should form a separate layer.
- Place the stopper in the neck of the funnel and gently agitate the contents of the funnel for a short while (put a finger on the stopper, invert, open the tap, agitate in a circular motion, close the tap and return the funnel to its normal position).
- Allow the contents to settle into two layers.
- Remove the stopper and open the tap to allow the lower layer to drain into a flask, and then do the same to allow the upper layer to drain into a separate flask.



▲ **fig A** Apparatus for separating immiscible liquids.

If a suitable solvent is used and the method is followed correctly, most of the desired organic product will have moved into the added solvent. It is better to use the solvent in small portions rather than in a single volume (e.g. four portions of 25 cm^3 rather than one portion of 100 cm^3) because this is more efficient. Using more portions of solvent, but with the same total volume, removes more of the desired organic product.

The desired organic product has been removed from the reaction mixture, but is now mixed with the added solvent. Therefore, simple distillation or fractional distillation must now be used to separate the desired organic product from the solvent used.

WASHING

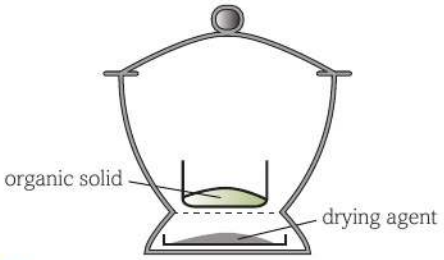
You might assume that washing is a technique used to remove impurities from a solid. Although this is correct, it is also used to remove impurities from a liquid. Washing could involve the use of water or an organic solvent. It could also involve the use of sodium carbonate solution, to remove any excess acid that may be present. Whichever liquid is used, it must be chosen carefully so as to dissolve the impurities but as little as possible of the substance being purified.

An impure solid is stirred in some of the solvent, then the mixture is filtered. If the solid is already in a filter funnel, the solvent could be added on top of the solid.

A liquid would be mixed with a solvent chosen so that it will dissolve little, if any, of the liquid to be purified. The mixture is then shaken in a separating funnel. After allowing the two liquid layers to separate, the tap is opened to allow each layer to drain into a separate container.

DRYING

No special technique is needed to dry an organic solid – it just needs to be left in a warm place or in a desiccator with a suitable drying agent (**fig B**).



▲ **fig B** Drying an organic solid in a desiccator

Many organic liquids are prepared using inorganic reagents, which are often used in aqueous solution. A liquid organic product may partially, or even completely, dissolve in water. Therefore, water may be an impurity that needs to be removed by a drying agent. Obviously, one important feature of a drying agent is that it does not react with the organic liquid.

Several drying agents are available, but the most common ones are anhydrous metal salts – usually calcium chloride, magnesium sulfate and sodium sulfate. The property that these compounds have in common is that they form hydrated salts. So, when they come into contact with water in an organic liquid, they absorb the water as water of crystallisation.

The drying agent is added to the organic liquid and the mixture is gently agitated or shaken, and then left for a period of time. Before use, a drying agent is powdery, but after absorbing water it looks more crystalline. If a bit more drying agent is added and remains powdery, then this is an indication that the liquid is dry. The liquid also goes from cloudy to clear when water is removed.

The drying agent is removed either by decantation (pouring the organic liquid off the solid drying agent) or by filtration.

FILTRATION

It is likely that after preparation, an organic solid will need to be filtered at some stage. This always happens as part of recrystallisation, so the two pieces of apparatus (Buchner and Hirsch funnels) that are normally used will be considered under the next heading. The use of a vacuum pump means that these methods are described as filtration under reduced pressure.

RECRYSTALLISATION

When an organic solid has been prepared, it is likely to need purification. A traditional way of removing impurities is the technique of recrystallisation. The principle behind this technique is that a solid compound is dissolved in a suitable solvent that can dissolve all or most of any impurities but very little of the compound being purified. The steps used in a typical purification are:

- Add the impure solid to a conical flask.
- Add some of the chosen solvent and warm until the mixture nears the boiling temperature of the mixture.

- If there is still some undissolved solid, add further solvent and warm until the mixture boils again.
- Continue adding further solvent and heating until all of the soluble solid has dissolved.
- If insoluble impurities are present, then hot filtration could be done using fluted filter paper in a heated funnel.
- Allow the liquid to cool until crystals of the organic solid have formed.
- More crystals can be obtained by cooling the solution below room temperature in an ice bath.
- The mixture is then filtered to remove soluble impurities using a Buchner funnel or a Hirsch funnel (**fig C**).
- The crystals are washed with a small amount of ice-cold solvent and then dried in a desiccator or warm oven.

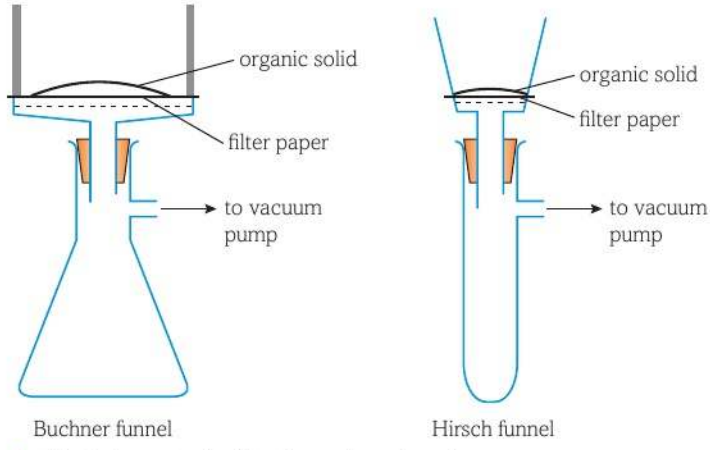
LEARNING TIP

The impure solid needs to be dissolved in the minimum volume of hot solvent. This is to make sure that the pure solid will crystallise when the solution is cooled.

Hot filtration is used to make sure that the pure solid does not crystallise at this stage.

Crystals of the pure solid form when the solution is cooled. This is because the solid is less soluble in the cold solvent than in the hot solvent.

The crystals are washed with a small amount of ice-cold solvent. This is to remove any soluble impurities, and to make sure that very little of the pure solid dissolves.



▲ **fig C** Apparatus for filtration under reduced pressure.

LEARNING TIP

Study the information about recrystallisation carefully. Many students do not understand why there might be two filtrations needed. Some throw away the substance they have just purified!

TESTING FOR PURITY

Having prepared an organic compound, a simple test can be performed to give an indication of whether the compound is pure. If the compound is a solid, then its melting temperature can be measured. If it is a liquid, then its boiling temperature can be measured.

LEARNING TIP

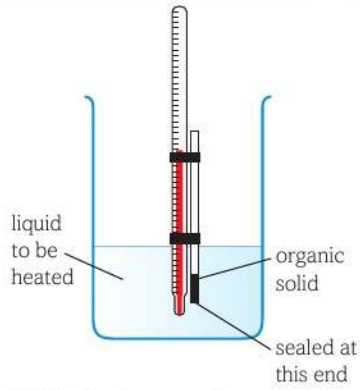
A pure solid has a sharp melting temperature rather than melting over a range of temperatures. This can be used as a test for purity.

DETERMINATION OF MELTING TEMPERATURE

For solids, impurities reduce the melting temperature. If you measure the melting temperature of your organic compound, then you can compare it with the known value of the pure compound. This will enable you to estimate how pure it is.

The traditional way to measure a melting temperature of a solid is to place some of the solid in a small capillary tube attached to the bulb of a thermometer and then place the assembly in a liquid that has a boiling temperature above that of the melting temperature of the solid (**fig D**).

In practice, this apparatus is often replaced by various electrical devices.



▲ **fig D** Apparatus for determination of melting temperature.

DETERMINATION OF BOILING TEMPERATURE

For liquids, impurities increase the boiling temperature.

The boiling temperatures of pure organic compounds have been carefully measured and are widely available in data books and online. If you measure the boiling temperature of your organic compound, you can compare the value you obtain with the known value of the pure compound and then estimate how pure the compound is.

The apparatus used depends on the volume of liquid available, and whether it is toxic or flammable. The apparatus used for simple distillation can be used in most cases.

A word of caution. This test may not be enough proof because you may not be able to measure the boiling temperature of your organic compound accurately enough – your thermometer might read too low or too high. So, even if your measured boiling temperature exactly matches the one in the data book or online, you may wrongly assume that your compound is pure.

It is also worth remembering that different organic compounds can, by coincidence, have the same boiling temperature. For example, both 1-chloropentane and 2-methylpropan-1-ol boil at 108 °C.

CHECKPOINT

1. Explain why the solvent extraction method will not work if the solvent dissolves the impurities.

2. Explain why, in recrystallisation, insoluble impurities are removed from a hot solution, not a cooled one.

SKILLS CRITICAL THINKING, SELF-DIRECTION